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Optical and electrical readout of diamond NV centers in dependence of the excitation wavelength

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ABSTRACT

Nitrogen-vacancy (NV) centers in diamond can be read out optically or electrically. To better understand the physics of their internal electronic transitions as well as ionization and recharging, we study the spin contrast observed in both readout methods as a function of the excitation wavelength for both single NV centers and ensembles. While the optical readout works best between 525 and 550 nm, the highest electrical spin contrast is achieved by resonantly exciting the NV^0 zero-phonon line at 575 nm. Additional spectral features attributed to a further excited state of NV^-, phonon sidebands and tunnel coupling to nitrogen donors are observed and discussed.

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The nitrogen-vacancy (NV) center in diamond is one of the quantum systems frequently used in practical applications. While single NV centers are especially relevant for quantum communication,[1] computing,[2,3] or sensing with high spatial resolution,[4] large ensembles offer a high signal intensity and sensitivity, as required for bulk magnetometry.[5] However, the widely applied optically detected magnetic resonance (ODMR) requires a complex setup and is ultimately limited by the inefficient outcoupling of photons.[6,7] Alternatively, also (photo) electrically detected magnetic resonance (EDMR or PDMR)[8] offers full spin readout,[9] nuclear spin detection,[10] and direct device implementation, for example, via all-diamond p-i-n devices.[11] A very powerful parameter to gain physical insight into the optical and electronic transitions of the NV center as well as to improve both readout methods is the excitation wavelength λ_{ex}.

Here, we apply optical and electrical spin contrast spectroscopy to investigate the NV center ionization and recharging cycle. We identify the optimal λ_{ex} to maximize the spin contrast in continuous wave (cw) EDMR and ODMR for ensembles and single NV centers in different diamond host materials. While both readout methods generally yield significant contrast between 480 and 580 nm, we additionally observe two resonant features in both the electrical and the optical readout around 521 and 575 nm, respectively. We attribute the first to enhanced ionization of NV^- via a resonant state in the conduction band[12,13] and the second to the zero-phonon line (ZPL) of NV^0, where especially efficient recharging to NV^-[14] leads to the highest EDMR contrast observed in this study.

The standard ODMR protocol is based on the spin-dependent photoluminescence (PL) of NV^- that originates from the transition between the ^3E excited and the $^3A_2$ ground triplet state [Fig. 1(a)]. This transition can be excited optically with $\lambda_{ex} \leq 637$ nm and features two different relaxation channels: a direct spin-conserving decay under the emission of a photon and a non-radiative intersystem crossing (ISC) to the $^1A_1$ singlet.[15] As the ISC is more likely for the $m_s = \pm 1$ sublevels of ^3E, these so-called dark states do not only produce less PL but also have a shorter lifetime than the bright $m_s = 0$. Consequently, also the photoionization probability of ^3E is spin-dependent, leading to a higher photocurrent (PC) from the long-lived $m_s = 0$ sublevel.[16] This two-step ionization from $^3A_2$ with ^3E as intermediate level shows a quadratic dependence on the excitation power P_{ex}[8,14] and carries the EDMR signal [Fig. 1(d)]. The direct ionization from $^3A_2$ to NV^0, on the other hand, is a single-photon process with an experimentally reported threshold of $\lambda_{ex} = 455$[17] or 475 nm[18] and is not expected to

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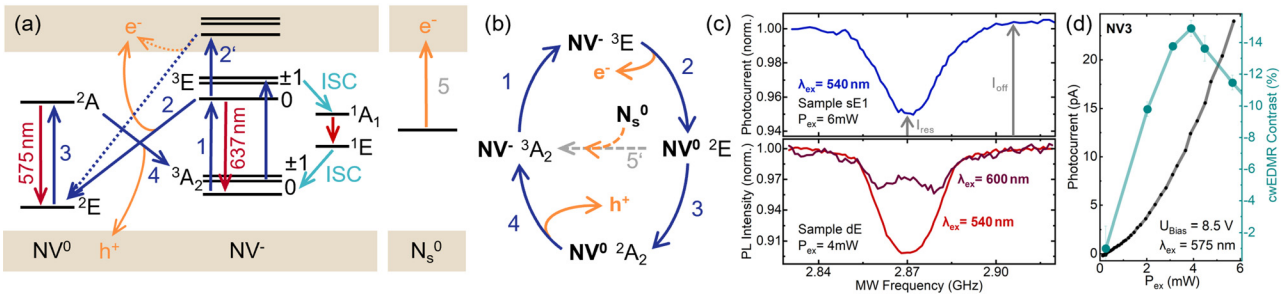


FIG. 1. (a) Photoionization and recharging of the NV center (transitions 1–4). Ionization is increased by resonant excitation into a further excited NV^- triplet state (transition 2′), which allows one electron to relax into the conduction band (dotted arrows). Non-radiative intersystem crossing (ISC) to the $^1A_1$ singlet (turquoise) is more likely from the $m_s = \pm 1$ levels of ^3E, leading to increased photoluminescence (PL, red) and photocurrent (orange) from $m_s = 0$. The ionization of N_s^0 (transition 5) contributes to the background current. (b) In the isolated charge state conversion cycle (CSCC, blue arrows) between NV^- and NV^0, transitions 2 and 4 create one free charge carrier each. Modifying the CSCC by electron exchange with N_s^0 (dashed arrows) results in a reduced generation of free carriers. (c) Representative cwEDMR (sample with low nitrogen concentration sE1, blue) and cwODMR (sample with high nitrogen concentration dE, red) spectra. Gray arrows mark the resonant I_{res} and off resonant photocurrent I_{off} as required for the calculation of the spin contrast. For the dE sample, excitation with wavelengths $\lambda_{ex} \geq 575$ nm (dark red) reveals a significant splitting arising from the Stark shift by nearby N_s^+. (d) Dependence of the photocurrent (black) and cwEDMR contrast (turquoise) for a single NV center on the optical excitation power P_{ex}. For each sample and excitation wavelength, P_{ex} was optimized for the highest spin contrast.

show any spin dependence. After ionization, electrons needed for the recharging can be excited from the valence band.[14,19] This requires two photons of $455 \leq \lambda_{ex} \leq 575$ nm[18] for the purely photoinduced NV charge state conversion cycle (CSCC) depicted in Fig. 1(b). As the whole CSCC depends on the ionization probability of ^3E, also the hole current generated in the recharging becomes spin-dependent. However, our results demonstrate that this isolated CSCC can be modified by nearby defects such as the nitrogen donor N_s^0, from where the exchange of an electron [transition 5′ in Fig. 1(b)] offers an alternative mechanism for the NV recharging. In addition, a single photon of $\lambda_{ex} \lesssim 563$ nm[20] can ionize N_s^0 and contribute directly to the spin-independent background PC [transition 5 in Fig. 1(a)].

We compare data from samples of different NV and N_s^0 concentrations grown by chemical vapor deposition (CVD) or high pressure–high temperature (HPHT) methods: (i) single NVs in three different ultra-pure type IIa CVD diamonds (ELSC plate with $[N_s^0] < 5$ ppb, element six) labeled NV1, NV2, and NV3, (ii) sparse ensembles in two different type IIa CVD diamonds (SC plate, element six) with $[NV] \leq 1$ ppb and $[N_s^0] \approx 170$ ppb labeled sE1 and sE2, and (iii) a high-density ensemble in electron-irradiated type Ib HPHT diamond (scimed), with $[NV] \approx 3.6$ ppm and $[N_s^0] \approx 95$ ppm, labeled dE. While the latter is a standard ODMR sample, we found that type IIa CVD diamond exhibits electronic properties favorable for EDMR. In all investigated samples, N_s^0 (determined by classical ESR and FTIR) is the dominant defect; however, we observe also small amounts of nitrogen-vacancy-hydrogen (NVH) center (by FTIR and UV-VIS spectroscopy) and H3 (by PL spectroscopy upon 480 nm excitation) in sE1 and sE2, while dE contains a significant H3 concentration. For PC measurements, we fabricate TiC electrodes of 10 μm distance on the diamond surface and apply a bias voltage of $U_{Bias} = 8.5$ V unless otherwise noted. A perpendicular bonding wire serves as microwave antenna to induce transitions between the $m_s = 0, \pm 1$ levels of $^3A_2$ at 2.87 GHz. The microwave power was adjusted to yield a π-pulse duration of 120 ns for each sample. PC measurements are performed analogously to Hrubesch *et al.*,[9] using a transimpedance amplifier (amplification $1\,\mathrm{GV\,A^{-1}}$) and a digital multimeter. We simultaneously detect the PL with a single-photon avalanche photodiode, protected by a 650 nm longpass filter. For optical excitation, we use a continuous wave laser (C-Wave VIS, HUBNER Photonics), tunable for wavelengths from 450 to 526 nm and 540 to 650 nm with a linewidth of 10 MHz, focused on a ≤ 1 μm spot. By scanning this spot over the diamond, we record spatial PL and PC maps (Fig. 2) and position the spot in the PC maximum before each measurement. In the ensemble samples, this maximum is always located directly next to the negatively biased electrode as expected in a metal–semiconductor–metal photoconductor.[22] The spin contrast C was measured without applying a magnetic field and is calculated from $C = (I_{res} - I_{off})/I_{off}$, with I_{res} and I_{off} being the PC or PL intensity upon resonant and off resonant microwave irradiation as depicted in Fig. 1(c). To maximize C for each tested wavelength and sample, we optimized the excitation power P_{ex} for EDMR and ODMR

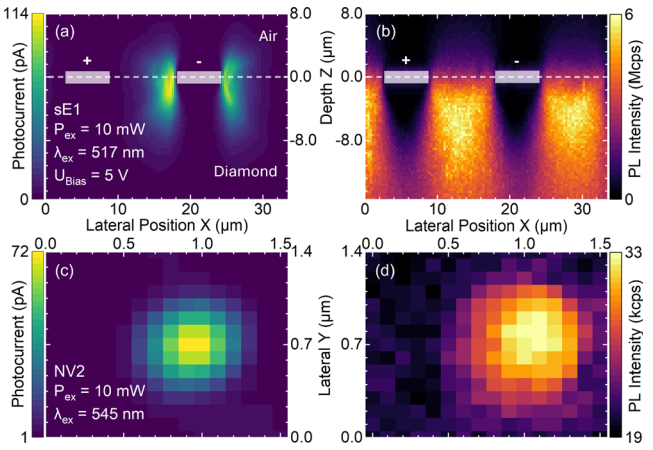


FIG. 2. (a) and (b) Cross-sectional photocurrent (PC) and photoluminescence (PL) maps of the sparse ensemble in sample sE1. PC originates from the vicinity of the negatively biased electrode (gray) close to the diamond surface (dashed line). (c) and (d) Lateral PC and PL maps of single NV2. Due to the two-step ionization process, the spatial FWHM is smaller in PC (0.46 μm) than in PL (0.78 μm), similar to Siyushev *et al.*[21]

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separately [cf. Fig. 1(d)]. For the EDMR measurements on NV2, a background photocurrent was subtracted from all data. We determined this background prior to each individual measurement by focusing the laser in 1.5 μm lateral distance to the single center. All measurements are performed at room temperature.

The reliable measurement of optical and electrical spin contrast in the isolated NV CSCC [Fig. 1(b)] requires the efficient driving of all four transitions, each with a distinct absorption cross section. To understand the interplay of these individual excitation processes, we first measure photoluminescence (PLE) and photocurrent (PCE) excitation spectra for all investigated samples. While our PLE spectra are consistent with previous studies,[12,18] no NV PCE data for comparison have been reported to date. In Fig. 3(a), we show PLE and PCE spectra for single NVs in direct comparison to the cwODMR [Fig. 3(b)] and cwEDMR [Fig. 3(c)] spectra of individual centers and ensembles. We discuss our findings by distinguishing three spectral regions of λ_{ex} that lead to different dynamics in the CSCC.

The first region with $\lambda_{ex} \leq 480$ nm is characterized by high PC, likely arising from the ionization of both the NV center and N_s^0. The PL intensity on the other hand remains comparatively low and is mainly attributed to NV^0, as evidenced by the wavelength-dependent PL ratio of both charge states in the inset of Fig. 3(a).[23] We attribute this to the increasing cross section of the one-photon ionization directly from the NV^- ground state $^3A_2$, which increases toward shorter λ_{ex} and replaces the successive transitions 1 and 2 in the CSCC [Fig. 1(b)]. However, direct ionization of NV^- prevents the effective population of the ^3E level, where the spin-dependent ISC to the $^1A_1$ singlet takes place. Therefore, PC and PL both depend only weakly on the NV^- spin state, and we observe low EDMR and ODMR contrast.

Upon optical excitation in the second spectral region between 480 and 575 nm, the ionization of NV^- now occurs in the two consecutive steps via the excited triplet state ^3E, forming the four-step CSCC as depicted in Fig. 1(b). This leads to high PLE and PCE rates in Fig. 3(a) and causes a significant increase in cwEDMR and cwODMR contrast for all samples. Both contrasts remain high over the entire regime of two-photon ionization up to the NV^0 ZPL at 575 nm, above which transition 3 can no longer be driven, and the conversion cycle breaks down. When exciting the ZPL of NV^0 resonantly, we find the highest electrical contrast of -15% for single NV centers and -22% for the sparse ensemble sE2 [Fig. 4(b)]. At 575 nm, the optical contrast exhibits a local maximum for all samples, but the broad global cwODMR maxima of -24% (NV1), -31% (sE1), and -12% (dE) were found between 525 and 550 nm. In cwEDMR, however, this broad maximum is not observed; instead, the electrically read spin contrast increases gradually from 480 to 575 nm. Similar to Hruby *et al.*,[24] we attribute this to the enhanced ionization cross section of N_s^0 at shorter λ_{ex},[17,20] resulting in a higher background current. A simple method to separate the photocurrent PC $(P_{ex}) = \alpha + \beta \cdot P_{ex} + \gamma \cdot P_{ex}^2$ into background and NV-related photocurrent is its power dependence.[8] While the linear term is associated with single-photon ionization of mainly N_s^0, the two-step ionization and recharging of the NV center is known to exhibit a quadratic power dependence. For the example of single NV3, the inset in Fig. 3(c) shows the ratio γ/β of quadratic to linear PC determined for P_{ex} below saturation. Consistently, this ratio increases strongly with increasing λ_{ex} and peaks at 575 nm, indicating a particularly large contribution of spin-contrast-carrying PC to the total photocurrent. Below 480 nm, where the

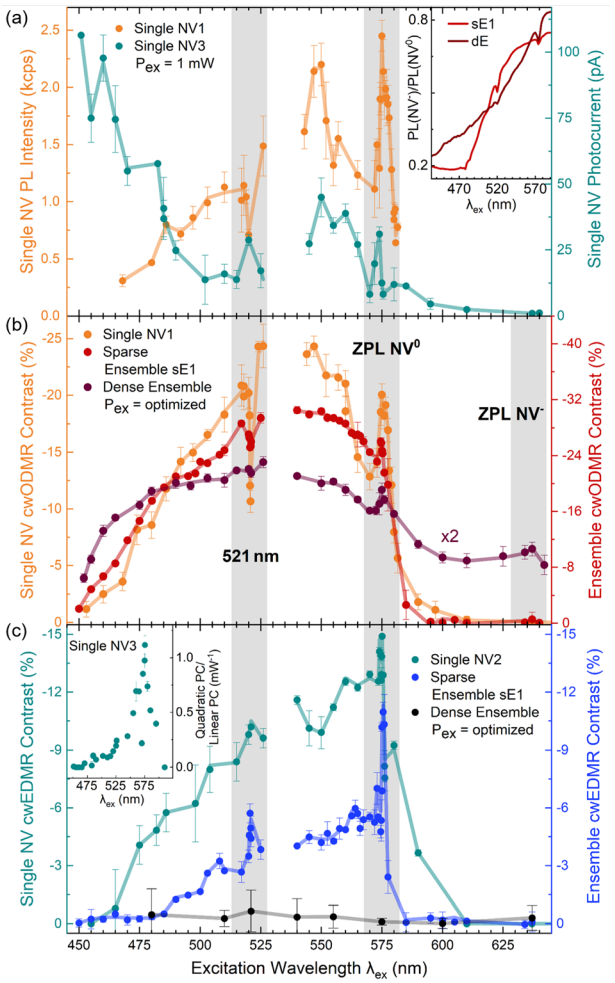


FIG. 3. (a) Continuous wave photoluminescence (PL) and photocurrent (PC) excitation spectra of single NV centers. PL is excited effectively only between 500 and 580 nm, whereas PC increases toward shorter excitation wavelengths λ_{ex}. The inset shows the λ_{ex}-dependent ratio of PL from NV^- and NV^0 for sE2 (bright) and dE (dark).[23] (b) cwODMR contrast in dependence of λ_{ex}, measured for single NV1 (orange) as well as a sparse (sE1, bright red) and a dense ensemble (dE, dark red). While the single NV and sE1 yield high cwODMR contrast between 480 and 580 nm, the contrast for dE is significantly smaller but extends over a much broader range up to the NV^- zero-phonon line (ZPL) at 637 nm. (c) λ_{ex}-Dependent cwEDMR contrast for NV2 (turquoise), sE1 (dark blue), and dE (black). Contrast is restricted to the same λ_{ex} as ODMR but decreases toward shorter λ_{ex} due to enhanced background PC. The inset shows the ratio of PC with a quadratic to PC with a linear power dependence, extracted from fitting PC(P_{ex}) with a second order polynomial. The gray shaded areas indicate the ZPLs of NV^0, NV^-, and the transition around 521 nm, attributed to a further excited triplet state of NV^-.[13] To improve visibility, data points are connected using B-spline interpolation (solid lines). Error bars show the standard deviation from 10 to 20 consecutive measurements. All measurements were performed at room temperature with an excitation laser linewidth of 10 MHz [except for the inset in (a) with $\Delta\lambda_{ex} \leq 3.0$ nm[23]]. While PLE and PCE are recorded at a fixed excitation power of 1 mW, P_{ex} in cwODMR and cwEDMR was optimized between 0.1 and 10 mW for each data point individually to yield the maximum spin contrast.

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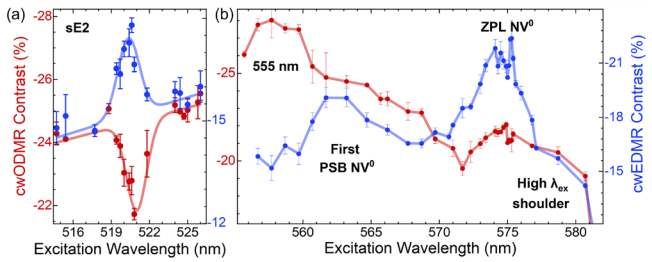


FIG. 4. cwEDMR (blue) and cwODMR (red) contrast of the sparse ensemble sE2. (a) cwEDMR maximum and cwODMR minimum around $\lambda_{ex} = 521$ nm, attributed to the increased ionization probability of ^{3}E due to a further NV$^-$ triplet state.[13] Solid lines are Gaussian fits to the data. (b) cwEDMR and cwODMR maxima for resonant excitation of the NV^0 ZPL, broadened due to locally varying lattice strain and accompanied by a high-wavelength shoulder. Additional maxima are found at the first NV^0 phonon sideband (PSB) in cwEMDR and at 555 nm in cwODMR. All measurements were performed at room temperature with an excitation laser linewidth of 10 MHz, and the excitation power was individually optimized for each data point to maximize the spin contrast.

ionization cross section of N_s^0 is high and also NV$^-$ is ionized by a single photon, the power dependence becomes exclusively linear.

In the third region $\lambda_{ex} > 575$ nm, we find a drastic decrease in both EDMR and ODMR contrast as well as in PLE and PCE efficiency of all samples except for dE. Generally, breaking transition 3 in the CSCC effectively pumps the NV center into its then dark, neutral charge state. The dE sample constitutes an exception in this respect, as significant cwODMR contrast is measured up to $\lambda_{ex} = 637$ nm, the ZPL of NV$^-$. We interpret this in terms of the high N_s^0 concentration, which can intervene with the NV CSCC in different ways. In addition to subsequent ionization and capture processes via the conduction band that can occur in all samples, the tunneling process studied by Manson *et al.*[25] can become relevant at small NV–N_s^0 distances as in our dE sample. The formation of NV$^-$–N_s^+ pairs via electron tunneling [transition 5′ in Fig. 1(b)] ensures a significant population of the negative NV charge state also for illumination with $\lambda_{ex} < 480$ nm or $\lambda_{ex} > 575$ nm [inset in Fig. 3(a)]. Then, at low excitation power, where ionization is negligible, spin-dependent PL can still be measured until transition 1 in the CSCC breaks down at 637 nm. This effect could be exploited to develop NV quantum sensors working with red excitation light as required for the investigation of materials that are photoactive under green illumination. However, it should be noted that the PL upon $\lambda_{ex} \geq 575$ nm originates exclusively from NV centers located in the immediate vicinity of a charged N_s^+, and the resulting Stark shift causes a significant splitting of the 2.87 GHz resonance, as shown in Fig. 1(c).[25] For cwEDMR, the high N_s^0 concentration and irradiation treatment appear less beneficial: In dE, the photocurrent exhibits a sublinear power dependence and shows a slow temporal response to changes in external parameters such as P_{ex} or U_{Bias}. This behavior is indicative of a highly defective diamond, where trapping and recombination processes have significant impact on the charge carrier dynamics.[26] Consequently, the EDMR contrast remains well below 1% over the entire investigated λ_{ex} range.

Within the three broad λ_{ex} regions, we observe at least two more narrow features. The first is a cwEDMR maximum and cwODMR minimum around $\lambda_{ex} = 521$ nm, as shown in more detail in Fig. 4(a) for the example of sE2. We interpret this as the resonant driving of the transition from ^{3}E into a recently confirmed NV$^-$ triplet state in the conduction band.[12,13] The resulting increase in the ^{3}E ionization rate is also consistent with the PLE minimum and PCE maximum in Fig. 3(a) as well as the shift of the equilibrium charge state toward NV^0 [inset in Fig. 3(a)] observed at this wavelength.

The second region investigated in detail encloses the ZPL of NV^0 at 575 nm [Fig. 4(b)], where we found the maximal cwEDMR contrast for single NVs and ensembles. We ascribe this to an exceptionally effective recharging process, as charge-state-sensitive PLE measurements in previous studies indicate a critical role of transition 3 in the CSCC.[12,14] An accelerated recharging shifts the equilibrium state toward $^3A_2$ in NV$^-$, which is sensitive to the spin manipulation and promotes both the electrical and the optical readout. However, the cwODMR at the same time suffers from an increased PL background of NV^0. For cwEDMR, on the other hand, an accelerated CSCC, in general, increases the ratio of NV-related to background PC, which explains the much stronger expression of this spectral feature in the electrical readout. Furthermore, a fast excitation of NV^0 in particular ensures that recharging by charge transfer from neighboring defects such as N_s^0 [transition 5′ in Fig. 1(b)] is suppressed. A long occupancy of the NV^0 ground state increases the probability of capturing a free electron instead of generating a free hole during recharging, which would lead to an effective quenching of the photocurrent and EDMR contrast. Note that the practical importance of this recharging mechanism is highly dependent on the specific lattice environment, and a similar process involving nearby acceptor defects is discussed by Bourgeois *et al.*[27] In cwEDMR, we are also able to resolve the first phonon sideband (PSB) of NV^0, centered around 563 nm in Fig. 4(b). The second PSB around 551 nm is hardly visible, though we observe a dominant peak in the PLE and PCE spectra in Fig. 3(a) at this wavelength. This feature is highly dependent on the excitation power and was previously reported by Aslam *et al.*[18] as part of a double-peak structure. At the spectral position reported for the second peak, we find a high-wavelength shoulder to the NV^0 ZPL extending up to 582 nm in all spin contrast and excitation spectra. Remarkably, this feature is generally absent in the emission spectra, and while Aslam *et al.* provide no definitive explanation of its physical origin, they suggest that additional vibrational modes or another electronically excited state may be involved in the excitation process. For different positions in the diamond sample, the ZPL itself shows a varying substructure in EDMR and ODMR, which is presumably related to lattice strain and Stark shift.[28–30]

Notably, Beha *et al.*[12] reported that 575 nm illumination is most efficient for exciting NV$^-$ photoluminescence, in particular at low temperature. Our experiments confirm this conclusion at room temperature under cw excitation and demonstrate that excitation at this wavelength leads to a strong local maximum in the cwODMR contrast and, most importantly, to a global maximum in the cwEDMR contrast. However, both studies, along with the strong temperature dependence of ODMR observed under green illumination,[31,32] motivate more detailed investigations of the wavelength dependence of optical and electrical readout as a function of temperature and excitation.

In summary, we studied the wavelength dependence of EDMR and ODMR for single NVs as well as sparse and high-density ensembles. By introducing optical and electrical spin contrast spectroscopy, we found three different spectral regions that drive different excitation dynamics in the NV charge state conversion cycle. For both readout

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methods, only little spin contrast is observed upon single-photon ionization of NV^- ($\lambda_{ex} \leq 480$ nm) and trapping in the neutral charge state ($\lambda_{ex} > 575$ nm). In between, the ODMR contrast exhibits a broad maximum from 525 to 550 nm. The EDMR contrast increases toward longer λ_{ex} as a consequence of the wavelength-dependent background current and shows a clear global maximum for resonant driving of the NV^0 ZPL at 575 nm. In addition, also resonant excitation of the 521 nm transition between NV^- ^3E and a further excited triplet state boosts the EDMR contrast. Our findings not only highlight the importance of the recharging process for the electrical readout but also motivate the use of 575 nm instead of the typically applied green laser in future NV EDMR applications such as fully integrated sensors. Our preliminary studies[33] reveal comparable results for pulsed readout methods, though with a more complex dependence on excitation power and dynamics. The identification of an optimal single excitation wavelength is also the first step toward the design of multicolored pulse sequences, which allow us to measure transition times in the CSCC selectively and can be tailored to specific application tasks, such as charge state pumping or spin-to-charge conversion. In addition to optimizing the readout contrast, we show that optical and electrical spin contrast spectroscopy is also suited for the investigation of electronic transitions, including those that are not photoluminescent or involve multi-stage processes and are, therefore, difficult to access with conventional spectroscopy methods. This technique is also transferable to other physical systems, such as color centers in SiC and hBN.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Lina M. Todenhagen: Data curation (equal); Formal analysis (equal); Investigation (lead); Methodology (equal); Visualization (equal); Writing – original draft (equal). **Martin S. Brandt:** Conceptualization (equal); Funding acquisition (equal); Investigation (supporting); Methodology (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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